HR Analytics Dashboard-SQL Query Documentation

Developed and validated HR Analytics dashboards using Excel, Power BI, and Tableau by performing SQL-based data verification and insight generation.

- **Create Table Query**

```
CREATE TABLE hrdata
(
emp_no INT8 PRIMARY KEY,
gender VARCHAR(50) NOT NULL,
marital_status VARCHAR(50),
age_band VARCHAR(50),
age INT8,
department VARCHAR(50),
education VARCHAR(50),
education_field VARCHAR(50),
job_role VARCHAR(50),
business_travel VARCHAR(50),
employee_count INT8,
attrition VARCHAR(50),
attrition_label VARCHAR(50),
job_satisfaction INT8,
active_employee INT8
);
```

- Import CSV File into MS SQL Server

```
COPY hrdata
FROM 'D:\hrdata.csv'
DELIMITER ','
CSV HEADER;
```

- <u>BASIC KPI VALIDATION QUERIE</u>
- Total Employee Count

```
SELECT
SUM(employee_count) AS employee_count
FROM hrdata;
```

- Attrition Count

```
SELECT
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition = 'Yes';
```

- Attrition Rate

```
SELECT
ROUND(
(
   (SELECT COUNT(attrition)
    FROM hrdata
    WHERE attrition = 'Yes')
    /
    SUM(employee_count)::numeric
) * 100, 2
) AS attrition_rate
```

```
FROM hrdata;
```

- Active Employees

```
SELECT
SUM(employee_count)-
(
  SELECT COUNT(attrition)
  FROM hrdata
  WHERE attrition = 'Yes'
) AS active_employees
FROM hrdata;
```

- Average Age of Employees

```
SELECT
ROUND(AVG(age),0) AS average_age
FROM hrdata;
```

- <u>GENDER ANALYSIS QUERIES</u>
- Attrition by Gender

```
SELECT
gender,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition = 'Yes'
GROUP BY gender
ORDER BY COUNT(attrition) DESC;
```

- Employee Distribution by Gender

```
SELECT
gender,
SUM(employee_count) AS total_employees
FROM hrdata
GROUP BY gender;
```

- Attrition Rate by Gender

```
SELECT
gender,
ROUND(
(
COUNT(CASE WHEN attrition='Yes' THEN 1 END)::numeric
/
COUNT(*)
) * 100,2
) AS attrition_rate
FROM hrdata
GROUP BY gender;
```

- <u>DEPARTMENT ANALYSIS QUERIES</u>
- Department Wise Attrition

```
SELECT
department,
COUNT(attrition) AS attrition_count,
ROUND(
(
```

```
COUNT(attrition)::numeric
/
(
SELECT COUNT(attrition)
FROM hrdata
WHERE attrition='Yes'
)
) * 100,2
) AS percentage
FROM hrdata
WHERE attrition='Yes'
GROUP BY department
ORDER BY COUNT(attrition) DESC;
```

- Department Wise Employee Count

```
SELECT
department,
SUM(employee_count) AS total_employees
FROM hrdata
GROUP BY department
ORDER BY total_employees DESC;
```

- Department with Highest Attrition

```
SELECT
department,
COUNT(*) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY department
ORDER BY attrition_count DESC
LIMIT 1;
```

- <u>AGE GROUP ANALYSIS</u>
- Number of Employees by Age

```
SELECT
age,
SUM(employee_count) AS employee_count
FROM hrdata
GROUP BY age
ORDER BY age;
```

- Attrition by Age Group

```
SELECT
age_band,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY age_band
ORDER BY attrition_count DESC;
```

- Attrition Rate by Age Group

```
SELECT
age_band,
ROUND(
```

```
(
COUNT(CASE WHEN attrition='Yes' THEN 1 END)::numeric
/
COUNT(*)
) * 100,2
) AS attrition_rate
FROM hrdata
GROUP BY age_band
ORDER BY attrition_rate DESC;
```

- <u>EDUCATION ANALYSIS</u>
- Education Wise Attrition

```
SELECT
education,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY education
ORDER BY attrition_count DESC;
```

- Education Field Wise Attrition

```
SELECT
education_field,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY education_field
ORDER BY COUNT(attrition) DESC;
```

- Education Field with Highest Attrition

```
SELECT
education_field,
COUNT(*) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY education_field
ORDER BY attrition_count DESC
LIMIT 1;
```

- <u>MARITAL STATUS ANALYSIS</u>
- Attrition by Marital Status

```
SELECT
marital_status,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY marital_status
ORDER BY attrition_count DESC;
```

- <u>JOB ROLE ANALYSIS</u>
- Attrition by Job Role

```
SELECT
job_role,
COUNT(attrition) AS attrition_count
FROM hrdata
```

```
WHERE attrition='Yes'
GROUP BY job_role
ORDER BY attrition_count DESC;
```

- Top 5 Job Roles with Highest Attrition

```
SELECT
job_role,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY job_role
ORDER BY attrition_count DESC
LIMIT 5;
```

- <u>JOB SATISFACTION ANALYSIS</u>
- **Job Satisfaction Rating by Job Role**

```
SELECT *
FROM crosstab(
'
SELECT
job_role,
job_satisfaction,
SUM(employee_count)
FROM hrdata
GROUP BY job_role, job_satisfaction
ORDER BY job_role, job_satisfaction
'
)
AS ct
(
job_role VARCHAR(50),
rating_1 NUMERIC,
rating_2 NUMERIC,
rating_3 NUMERIC,
rating_4 NUMERIC
)
ORDER BY job_role;
```

- Average Job Satisfaction by Department

```
SELECT
department,
ROUND(AVG(job_satisfaction),2) AS avg_job_satisfaction
FROM hrdata
GROUP BY department
ORDER BY avg_job_satisfaction DESC;
```

- Business Travel Impact on Attrition

```
SELECT
business_travel,
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY business_travel
ORDER BY attrition_count DESC;
```

- Most Common Age Group in Company

```
SELECT
age_band,
COUNT(*) AS employee_count
FROM hrdata
GROUP BY age_band
ORDER BY employee_count DESC
LIMIT 1;
```

- Gender Distribution Across Departments

```
SELECT
department,
gender,
COUNT(*) AS employee_count
FROM hrdata
GROUP BY department, gender
ORDER BY department;
```

- Attrition Percentage by Education Field

```
SELECT
education_field,
ROUND(
(
COUNT(CASE WHEN attrition='Yes' THEN 1 END)::numeric
/
COUNT(*)
) * 100,2
) AS attrition_percentage
FROM hrdata
GROUP BY education_field
ORDER BY attrition_percentage DESC;
```

- Employees with High Job Satisfaction

```
SELECT
job_role,
COUNT(*) AS employee_count
FROM hrdata
WHERE job_satisfaction >= 3
GROUP BY job_role
ORDER BY employee_count DESC;
```

- Employees with Low Job Satisfaction

```
SELECT
job_role,
COUNT(*) AS employee_count
FROM hrdata
WHERE job_satisfaction <= 2
GROUP BY job_role
ORDER BY employee_count DESC;
```

- Attrition vs Job Satisfaction

```
SELECT
job_satisfaction,
```

```
COUNT(attrition) AS attrition_count
FROM hrdata
WHERE attrition='Yes'
GROUP BY job_satisfaction
ORDER BY job_satisfaction;
```

- Department Having Maximum Active Employees

```
SELECT
department,
COUNT(*) AS active_employees
FROM hrdata
WHERE attrition='No'
GROUP BY department
ORDER BY active_employees DESC
LIMIT 1;
```

- Attrition Contribution by Each Department

```
SELECT
department,
ROUND(
(
COUNT(*)::numeric
/
(SELECT COUNT(*)
FROM hrdata
WHERE attrition='Yes'
)
) * 100,2
) AS attrition_contribution_percentage
FROM hrdata
WHERE attrition='Yes'
GROUP BY department
ORDER BY attrition_contribution_percentage DESC;
```

- Age Group Wise Gender Attrition

```
SELECT
age_band,
gender,
COUNT(attrition) AS attrition_count,
ROUND(
(
COUNT(attrition)::numeric
/
(
SELECT COUNT(attrition)
FROM hrdata
WHERE attrition='Yes'
)
) * 100,2
) AS percentage
FROM hrdata
WHERE attrition='Yes'
GROUP BY age_band, gender
ORDER BY age_band, gender DESC;
```